Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/21/2005
PI Name:	Delp, Michael Ph.D.		
Project Title:	Adaptations of Cerebral Arteries to Simulated Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	32306-0001	**Congressional District:**	2
Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	03/01/2004	**End Date:**	08/31/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Changed end dateof 9/30/2006.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CC66G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	These studies will address the relatively unexplored question of the effects of simulated microgravity on vascular smooth muscle and endothelial cell function in resistance arteries isolated from splanchnic, renal and cerebral tissues. Alterations in vascular function induced by microgravity are particularly relevant to the problems of orthostatic intolerance experienced by astronauts upon re-entry into the earth's gravitational field. Previous research has demonstrated that reductions in blood volume and altered baroreceptor reflex control of the cardiovascular system contribute to, but do not fully account for, the changes in orthostatic tolerance. The problem of altered resistance vessel reactivity and morphology will be studied using a ground-based model to simulate a microgravity environment, i.e., the tail-suspended, hindlimb-unloaded rat. The impact of simulated microgravity on maximal responsiveness and sensitivity of resistance arteries to a variety of vasoconstrictor and vasodilator agents, each acting through different mechanisms, will be determined. This work will provide potentially important information about the mechanism(s) underlying the orthostatic intolerance of astronauts returning to Earth. Such information would be useful in the development of more effective countermeasures. The impact of simulated microgravity on maximal responsiveness and sensitivity of resistance arteries to a variety of vasoconstrictor and vasodilator agents, each acting through different mechanisms, will be determined. This work will provide potentially important information about the mechanism(s) underlying the orthostatic intolerance of astronauts returning to Earth. Such information would be useful in the development of more effective countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These experiments will expand our understanding of how prolonged fluid shifts, such as in a microgravity environment or during bedrest, affect the cardiovascular system. The detailed study of the resulting vascular adaptations triggered by simulated microgravity will yield essential information on the basic physiological responses of individual blood vessels involved in blood flow and pressure regulation. This information will also support the development of treatments or countermeasures to improve crew health and performance following their return to a gravitational environment.
Task Progress:	Aim 1, to determine whether head down tilt (HDT) in rodents increases in middle cerebral artery (MCA) spontaneous tone, as well as myogenic, shear stress and KCl-evoked vasoconstriction are the result of defective vasodilator signaling mechanisms has been addressed. The results are in press in the American Journal of Physiology-Heart and Circulatory Physiology. Aim 2, to determine whether the HDT-induced smooth muscle hypertrophy and increases in medial wall thickness alters the mechanical properties of MCAs, has not been addressed. Experiments described in Aim 3, to determine whether endothelium-dependent vasodilation of MCAs is diminished by HDT, are in progress. Aim 4, to determine whether alterations in cerebral resistance artery spontaneous tone and vasoconstrictor responsiveness (see Aim 1), as well as putative alterations in vasodilator responsiveness and mechanical properties described in vitro function to diminish cerebral perfusion and elevate cerebral vascular resistance in vivo have been addressed. The results are in press in the American Journal of Physiology-Heart and Circulatory Physiology.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Abstracts for Journals and Proceedings	Delp MD, Wilkerson MK. "Simulated microgravity diminishes regional cerebral perfusion in rats." Aerospace Medical Association Annual Meeting, May 2004. Aviat Space Environ Med (Suppl) 2004;75:B116-B117. , May-2004